

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XI: THE PGL_3 POSITIVE HECKE-DYNAMICAL SYSTEM: SHIFT EQUIVALENCE OVER $\mathbb{Z}$ AND THE GEODESIC EDGE FLOW

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ABSTRACT. Paper X left one problem at the top of its list: a non-negative integer matrix shift equivalent over $\mathbb{Z}$ to the companion $C^{(3)}$ of the PGL_3 cubic pencil, which would be a positive Hecke-dynamical system carrying the Hecke critical group $\mathbb{Z} \oplus \mathrm{Jac}^H(Y)$ in its K -theory. This paper settles what can be settled about that problem and builds the positive system that does exist. First, the $\mathbb{Z}/3$ grading $\deg(i, v) = i - \tau(v)$ makes $C^{(3)}$ a period-three cyclic matrix; shift equivalence of $C^{(3)}$ to a non-negative matrix reduces, with an explicit lag formula, to shift equivalence of the return map $P_0 = C^{(3)3}|_{\deg 0}$ (Perron root q^6 , $\det P_0 = q^{3n}$) to a primitive matrix. Second, the required statement is identified precisely: since the nonzero spectrum of P_0 satisfies the Perron and net-trace conditions ([X], Kim–Ormes–Roush), its shift equivalence over $\mathbb{Z}$ to a primitive matrix is an instance of the *weak Generalized Spectral Conjecture* of Boyle–Handelman, which is open over $\mathbb{Z}$; the specification’s request for explicit R, S, ℓ is a request to prove a case of an open conjecture, and we do not claim to. Third, the natural explicit route — a positive presentation $(tI - Y)(tI - P_0)$ with a nilpotent cofactor Y — is proved impossible on Y_{21} and Y_{24} by exact rational Farkas certificates: P_0 has a positive right but not left Perron vector, no positive power, and a negative-entry digraph with cycles. Fourth, and positively: the *geodesic edge flow* L_E on typed edges $((x, y) \rightarrow (y, z)$ with z not adjacent to x) is a non-negative integer matrix with row sums q^2 , irreducible, and for every joint eigenvector f of (A_1, A_2) with eigenvalues (a, b) the three edge functions $f \circ \text{tail}$, $f \circ \text{head}$, m_f (sum over the third vertices of the chambers on the edge) span an L_E -invariant space on which the characteristic polynomial is exactly $\lambda^3 - a\lambda^2 + qb\lambda - q^3$. Hence an explicit integer intertwiner Ψ with $L_E\Psi = \Psi C'$, C' lag-one shift equivalent to $C^{(3)}$, and the whole nontrivial PGL_3 Hecke spectrum inside the spectrum of a non-negative matrix; the trivial pair contributes only q^2 ($\mathrm{rk} \Psi = 3n - 6$), and the complementary spectrum has modulus $q^{1/2}$ (numeric on Y_{168}). The Cuntz–Krieger algebra $\mathcal{O}_{L_E}$ is a Kirchberg algebra whose gauge dynamics is the gallery flow and whose K_0 is finite: the positive system sees the cuspidal Hecke spectrum but not the Hecke critical group, the rank-two form of the bulk–boundary dichotomy of [I].

STATUS OF THE RESULTS

Every numbered statement is proved except where marked “numeric” (floating-point spectra, Table 1). Theorem 2.3 rests on exact rational Farkas certificates verified by integer arithmetic. The classical inputs are the module characterization of shift equivalence over $\mathbb{Z}$ [3, §2], the Spectral Conjecture over $\mathbb{Z}$ [5], and the statements (not proofs) of the Generalized Spectral Conjecture [1, 2, 3]. Kang–Li [4] is context for the edge flow. The complexes are those of [X]; the edge-flow theorem uses the flag condition [X, Def. 1.1(iv)], which holds for Y_{168} and fails for the smaller covers, so its numerical checks are on Y_{168} . Section 5 records what is open and the corrections to the specification; Proposition 3.8 answers its item on KMS states.

Date: 18 September 2026.

2020 Mathematics Subject Classification. Primary 37B10, 15B48; Secondary 46L80, 20E42, 11F70, 05E45.

Key words and phrases. shift equivalence, primitive matrix, Generalized Spectral Conjecture, $\tilde{A}_2$ complex, Hecke operator, companion matrix, geodesic flow, Cuntz–Krieger algebra.

Paper XI of the series; Papers I–X are cited as [I]–[X]. The verification script `pgl3_dynamical.py` in the ancillary bundle reproduces every computational claim; its output is archived as `pgl3_dynamical_output.txt`.

1. THE GRADING, THE RETURN MAP, AND THE CYCLIC REDUCTION

Throughout, Y is an $\tilde{A}_2$ complex of order q in the sense of [X, Def. 1.1] with n vertices, type function τ , type-raising adjacency A_1 , $A_2 = A_1^t$, and companion $C^{(3)} = \begin{pmatrix} A_1 & -qA_2 & q^3I \\ I & 0 & I \\ 0 & I & 0 \end{pmatrix}$ on $\mathbb{Z}^{3n} = \bigoplus_{i=0}^2 \mathbb{Z}^V$, $\det(I - uC^{(3)}) = \det P(u)$ with $P(u) = I - uA_1 + qu^2A_2 - q^3u^3I$ [X, Thm. 3.3].

Lemma 1.1 (Grading). *Give the coordinate (i, v) of $\mathbb{Z}^{3n}$ the degree $\deg(i, v) := i - \tau(v) \in \mathbb{Z}/3$. Then $C^{(3)}$ maps the degree- d summand $\mathbb{Z}_d^{3n}$ into $\mathbb{Z}_{d+1}^{3n}$: in the graded decomposition $C^{(3)}$ is the cyclic block matrix with blocks $X_d: \mathbb{Z}_d^{3n} \rightarrow \mathbb{Z}_{d+1}^{3n}$, each $\mathbb{Z}_d^{3n}$ of rank n , and $C^{(3)3}$ is block diagonal with return maps $P_d := X_{d+2}X_{d+1}X_d \in M_n(\mathbb{Z})$.*

Proof. A_1 sends $(0, y)$ to $(0, x)$ with $\tau(x) = \tau(y) - 1$, raising the degree by one; the identity blocks send $(0, v) \mapsto (1, v)$ and $(1, v) \mapsto (2, v)$, raising i by one; $-qA_2$ sends $(1, v)$ to $(0, x)$ with $\tau(x) = \tau(v) + 1$: degree $1 - \tau(v) \mapsto -\tau(v) - 1 \equiv 1 - \tau(v) + 1$; q^3I sends $(2, v) \mapsto (0, v)$: $2 - \tau(v) \mapsto -\tau(v) \equiv 3 - \tau(v)$. Each $\mathbb{Z}_d^{3n}$ has rank n because the n vertices split equally among the three types [X, Lem. 1.2]. Verified as an exact block identity on four complexes (Table 1, line (G)). $\square$

Proposition 1.2 (The return map). *The nonzero spectrum of P_0 is Λ_Q of [X, Def. 4.4]: one cube λ^3 for each ω -orbit of eigenvalues of $C^{(3)}$, with q^6 simple and dominant; $\det P_0 = q^{3n}$; P_0 has a positive right Perron vector, namely the restriction of $(q^4\mathbf{1}, q^2\mathbf{1}, \mathbf{1})$ to $\mathbb{Z}_0^{3n}$. On $Y_{21}, Y_{24}, Y_{42}, Y_{168}$ (numeric): the left Perron vector of P_0 is not positive, no power P_0^k , $k \leq 30$, is positive, and the digraph of negative entries of P_0 contains cycles.*

Proof. $C^{(3)3} = \bigoplus_d P_d$ and the three P_d are conjugate over $\mathbb{Z}$ ($P_{d+1}X_d = X_dP_d$ with X_d invertible over $\mathbb{Q}$), so each has the spectrum of $C^{(3)3}$ with multiplicities divided by three: $\{\lambda^3\}$, one per orbit. $\det C^{(3)} = (q^3)^n$ (block expansion), so $\det P_0^3 = \det C^{(3)3} = q^{9n}$ and $\det P_0 = q^{3n}$ (positive, as q^6 dominates). For $\lambda = q^2$ the vector $(\lambda^2\mathbf{1}, \lambda\mathbf{1}, \mathbf{1})$ is a $C^{(3)}$ -eigenvector because $\lambda^2(q^2 + q + 1) - q\lambda(q^2 + q + 1) + q^3 = \lambda^3$; it is positive and graded-homogeneous pieces of it are eigenvectors of the P_d . The remaining statements are computations. $\square$

Lemma 1.3 (Cyclic lifting of shift equivalence). *Let $B_0 \in M_N(\mathbb{Z})$ and suppose (R_0, S_0, ℓ) is a shift equivalence over $\mathbb{Z}$ between P_0 and B_0 : $P_0R_0 = R_0B_0$, $S_0P_0 = B_0S_0$, $R_0S_0 = P_0^\ell$, $S_0R_0 = B_0^\ell$. Put $B := \text{cyc}(I, I, B_0)$, the cyclic block matrix with blocks $I: 0 \rightarrow 1$, $I: 1 \rightarrow 2$, $B_0: 2 \rightarrow 0$ on $\mathbb{Z}^N \oplus \mathbb{Z}^N \oplus \mathbb{Z}^N$, and*

$$R := \text{diag}(R_0, X_0R_0, X_1X_0R_0), \quad S := \text{diag}(S_2X_1X_0, S_2X_1, S_2), \quad S_2 := S_0X_2.$$

Then $C^{(3)}R = RB$, $SC^{(3)} = BS$, $RS = C^{(3)3(\ell+1)}$, $SR = B^{3(\ell+1)}$: a shift equivalence over $\mathbb{Z}$ of lag $3(\ell+1)$. If $B_0 \geq 0$ then $B \geq 0$, and B is irreducible of period 3 when B_0 is primitive. Conversely a shift equivalence between $C^{(3)}$ and a graded cyclic B restricts to one between the return maps.

Proof. Direct verification: $X_0R_0 = (X_0R_0) \cdot I$, $X_1(X_0R_0) = (X_1X_0R_0) \cdot I$, $X_2(X_1X_0R_0) = P_0R_0 = R_0B_0$ give $C^{(3)}R = RB$; $S_2X_1X_0 \cdot X_2 = S_0P_0X_2 = B_0S_0X_2 = B_0S_2$ and the two identities $S_{d+1}X_d = S_d$ give $SC^{(3)} = BS$; on the degree- d block RS equals $R_dS_d = (X_{d-1} \cdots X_0R_0)(S_0X_2 \cdots X_d) = X_{d-1} \cdots X_0P_0^\ell X_2 \cdots X_d = P_d^{\ell+1}$, and SR equals $S_0P_0R_0 = B_0^{\ell+1}$ on each block, while $B^3 = \text{diag}(B_0, B_0, B_0)$. Checked with the trivial equivalence $B_0 = P_0$ on four complexes (Table 1, line (L)). The converse is the restriction of the four equations to graded pieces. $\square$

2. SHIFT EQUIVALENCE OVER $\mathbb{Z}$: CRITERION, COFACTORS, OBSTRUCTIONS

Proposition 2.1 (Module criterion). *For square integer matrices A, B (any sizes), $A \sim_{\mathbb{Z}} B$ if and only if the $\mathbb{Z}[t, t^{-1}]$ -modules $M_A := \mathbb{Z}[t^\pm]^n / (tI - A)\mathbb{Z}[t^\pm]^n$ and M_B are isomorphic [3, §2]. For nonsingular A , $M_A \cong \mathbb{Z}^n[A^{-1}] = \bigcup_k A^{-k}\mathbb{Z}^n$ with t acting as A . A non-negative B with*

$B \sim_{\mathbb{Z}} P_0$ is therefore the same thing as a finite set of generators $g_1, \dots, g_N$ of M_{P_0} over $\mathbb{Z}[t^{\pm}]$ with $t g_j = \sum_i B_{ij} g_i$, $B_{ij} \in \mathbb{Z}_{\geq 0}$: a positive presentation of the Hecke module.

Proposition 2.2 (Cofactor construction). *Let $U(t) \in GL_n(\mathbb{Z}[t^{\pm}])$ and suppose $F(t) := U(t)(tI - P_0) = t^{K+1}I - \sum_{k=0}^K C_k t^k$ with all $C_k \in M_n(\mathbb{Z}_{\geq 0})$. Then the companion $B_F = \begin{pmatrix} C_K & C_{K-1} & \dots & C_0 \\ I & 0 & \dots & 0 \\ & \ddots & & \\ 0 & \dots & I & 0 \end{pmatrix} \geq 0$ satisfies $B_F \sim_{\mathbb{Z}} P_0$. In particular, if $Y \in M_n(\mathbb{Z})$ is nilpotent with $P_0 + Y \geq 0$ and $-Y P_0 \geq 0$, then $U = tI - Y$ works and $B = \begin{pmatrix} P_0 + Y & -Y P_0 \\ I & 0 \end{pmatrix} \geq 0$ is shift equivalent over $\mathbb{Z}$ to P_0 , hence $\text{cyc}(I, I, B) \sim_{\mathbb{Z}} C^{(3)}$ by Lemma 1.3.*

Proof. $\text{coker}(tI - B_F) \cong \text{coker } F$ by the companion reduction (as in [X, Thm. 3.3]), and $\text{coker } F = \text{coker}(U(tI - P_0)) \cong \text{coker}(tI - P_0)$ because U is invertible over $\mathbb{Z}[t^{\pm}]$; Proposition 2.1. For $U = tI - Y$: $\det(tI - Y) = t^n$ is a unit in $\mathbb{Z}[t^{\pm}]$ when Y is nilpotent, and $(tI - Y)(tI - P_0) = t^2 I - (P_0 + Y)t + Y P_0$. $\square$

Theorem 2.3 (No nilpotent cofactor on Y_{21}, Y_{24}). *For $Y = Y_{21}$ and $Y = Y_{24}$ there is no matrix $Y \in M_n(\mathbb{Q})$ with $P_0 + Y \geq 0$, $Y P_0 \leq 0$ and $\text{Tr } Y = 0$; a fortiori no nilpotent integer cofactor exists, and Proposition 2.2 with $K = 1$ cannot produce a positive presentation.*

Proof. Write $Y = -P_0 + Z$, $Z \geq 0$; the system becomes $Z P_0 \leq P_0^2$ entrywise and $\text{Tr } Z = \text{Tr } P_0$. By Farkas' lemma it is infeasible iff there exist $\Lambda \in M_n(\mathbb{Q}_{\geq 0})$ and $\mu \in \mathbb{Q}$ with $\Lambda P_0^t + \mu I \geq 0$ entrywise and $\langle \Lambda, P_0^2 \rangle + \mu \text{Tr } P_0 < 0$. The script finds such certificates with denominators 2 (Y_{21} , dual value -4340) and 128 (Y_{24} , dual value $-53185/8$) and verifies both inequalities in exact rational arithmetic. (For Y_{42} the LP relaxation is infeasible in floating point but a rational certificate was not extracted; we claim nothing there.) $\square$

Remark 2.4 (The precise status of the realization problem). Let Λ be the nonzero spectrum of P_0 . By [X, Thm. 4.5] and [5], Λ is the nonzero spectrum of a primitive integer matrix. The statement the specification asks for — $P_0 \sim_{\mathbb{Z}} B_0$ with B_0 primitive, equivalently $C^{(3)} \sim_{\mathbb{Z}} B$ with $B \geq 0$ (Lemma 1.3) — is exactly the *weak form of the Generalized Spectral Conjecture* [1, p. 253], [2, p. 124] for the ring $\mathbb{Z}$ and the matrix P_0 : that the three spectral conditions suffice for shift equivalence over $\mathbb{Z}$ to a primitive matrix. Boyle–Schmieding [3] show that over dense subrings of $\mathbb{R}$ the weak form implies the strong form (strong shift equivalence); $\mathbb{Z}$ is not dense and the weak form over $\mathbb{Z}$ is, to our knowledge, open. The requested explicit (B, R, S, ℓ) would therefore prove a case of an open conjecture. This paper does not. What it proves is that the problem is equivalent to the primitive-matrix problem for the $n \times n$ return map (Lemma 1.3), that the first explicit route is closed (Theorem 2.3), and that a positive system carrying the Hecke spectrum — though not the Hecke module — exists and is canonical (Section 3).

3. THE GEODESIC EDGE FLOW

Let E be the set of typed edges $e = (x \rightarrow y)$, $\tau(y) = \tau(x) + 1$, and for each chamber $\{x, y, z\}$ on e call z its third vertex. Let $S, T, M \in M_{n \times |E|}(\mathbb{Z})$ be the tail, head and third-vertex incidence matrices: $S_{v,e} = [v = \text{tail}(e)]$, $T_{v,e} = [v = \text{head}(e)]$, $M_{v,e} = [\{v\} \cup e \text{ is a chamber}]$; M has column sums $q + 1$. For a vertex function f put $f \circ t := S^t f$, $f \circ h := T^t f$, $m_f := M^t f$ (the sum of f over the third vertices of the chambers on the edge).

Definition 3.1 (Geodesic edge flow). $L_E \in M_{|E|}(\{0, 1\})$, $L_E((x \rightarrow y), (y \rightarrow z)) = 1$ iff $\tau(z) = \tau(y) + 1$ and z is not adjacent to x ; all other entries 0. A step of L_E extends a type-raising edge to a type-raising path of length two that does not lie in a chamber: the rank-two analogue of the non-backtracking condition of the Hashimoto matrix A^{nb} of [I].

Lemma 3.2 (Local counts). *Let Y satisfy [X, Def. 1.1(i)–(iv)] and have vertex links isomorphic to the incidence graph of $\text{PG}(2, q)$. For a typed edge $e = (x \rightarrow y)$:*

- (a) y has $q^2 + q + 1$ type-raising neighbours z ; exactly $q + 1$ of them are adjacent to x , and those are exactly the third vertices of the chambers on e (flag condition); hence the row sum of L_E at e is q^2 ;
- (b) $\sum_{z: y \rightarrow z, z \not\sim x} f(z) = (A_1 f)(y) - m_f(e)$;
- (c) $\sum_{z: y \rightarrow z, z \not\sim x} m_f(y \rightarrow z) = q((A_2 f)(y) - f(x))$.

Proof. (a),(b) are the flag condition. (c) Expand $m_f(y \rightarrow z) = \sum_w f(w)$ over third vertices w of chambers $\{y, z, w\}$; such w are type- $\tau(x)$ neighbours of y , i.e. the in-neighbours $w \rightarrow y$ counted by A_2 . For fixed w the number of z with $y \rightarrow z$, $\{y, z, w\}$ a chamber and $z \not\sim x$ is: 0 if $w = x$ (every chamber on $(x \rightarrow y)$ has $z \sim x$); and if $w \neq x$, in the link of y — the incidence graph of $\text{PG}(2, q)$, with x, w two distinct points and the z 's the $q + 1$ lines through w — the lines through w that also pass through x number exactly one, so q of them have $z \not\sim x$. Summing, $q \sum_{w \rightarrow y, w \neq x} f(w) = q((A_2 f)(y) - f(x))$. $\square$

Theorem 3.3 (The edge flow carries the Hecke pencil). *Under the hypotheses of Lemma 3.2, with $\Psi := \begin{bmatrix} S^t & T^t & M^t \end{bmatrix} \in M_{|E| \times 3n}(\mathbb{Z})$ and*

$$C' := \begin{pmatrix} 0 & 0 & -qI \\ q^2I & A_1 & qA_2 \\ 0 & -I & 0 \end{pmatrix} \in M_{3n}(\mathbb{Z}) : \quad L_E \Psi = \Psi C'.$$

Consequently, for every joint eigenvector f of (A_1, A_2) with eigenvalues (a, b) , the space $V_f := \text{span}\{f \circ t, f \circ h, m_f\}$ is L_E -invariant and the matrix of L_E on the ordered basis $(f \circ t, f \circ h, m_f)$ is $\begin{pmatrix} 0 & 0 & -q \\ q^2 & a & qb \\ 0 & -1 & 0 \end{pmatrix}$, with characteristic polynomial $\lambda^3 - a\lambda^2 + qb\lambda - q^3$, the cubic pencil of the pair (a, b) . Every root of the cubic pencil attached to a joint eigenpair with $\dim V_f = 3$ is an eigenvalue of the non-negative matrix L_E .

Proof. For $g = \alpha f \circ t + \beta f \circ h + \gamma m_f$ and $e = (x \rightarrow y)$,

$$\begin{aligned} (L_E g)(e) &= \sum_{z: y \rightarrow z, z \not\sim x} (\alpha f(y) + \beta f(z) + \gamma m_f(y \rightarrow z)) \\ &= \alpha q^2 f(y) + \beta((A_1 f)(y) - m_f(e)) + \gamma q((A_2 f)(y) - f(x)) \end{aligned}$$

by Lemma 3.2, i.e. $L_E g = (-q\gamma) f \circ t + (q^2\alpha f + \beta A_1 f + q\gamma A_2 f) \circ h + (-\beta) m_f$. Read for arbitrary (not necessarily eigen-) triples (f_0, f_1, f_2) this is $L_E \Psi(f_0, f_1, f_2)^t = \Psi(-qf_2, q^2f_0 + A_1f_1 + qA_2f_2, -f_1)^t = \Psi C'(f_0, f_1, f_2)^t$. For an eigenvector, $A_1 f = af$, $A_2 f = bf$ give the 3×3 matrix; expanding its characteristic determinant along the first row gives $\lambda((\lambda - a)\lambda + qb) - q^3$. Verified as an exact integer identity $L_E \Psi = \Psi C'$ on Y_{168} (Table 1, line (I)). $\square$

Proposition 3.4 (C' is the companion). *With $D := \begin{pmatrix} 0 & 0 & qI \\ I & 0 & 0 \\ 0 & -I & 0 \end{pmatrix}$, both D and $S := C^{(3)}D^{-1}$ are integer matrices, and $C'D = DC^{(3)}$, $SC' = C^{(3)}S$, $DS = C'$, $SD = C^{(3)}$: C' and $C^{(3)}$ are shift equivalent over $\mathbb{Z}$ with lag one (indeed elementary strong shift equivalent over $\mathbb{Z}$), so $M_{C'} \cong M_{C^{(3)}}$ and C' has the spectrum of the pencil.*

Proof. $D^{-1} = \begin{pmatrix} 0 & I & 0 \\ 0 & 0 & -I \\ q^{-1}I & 0 & 0 \end{pmatrix}$, and the third block column of $C^{(3)}$ is $(q^3I, 0, 0)^t$, so $C^{(3)}D^{-1}$ has first block column $(q^2I, 0, 0)^t$ and is integral. The four identities are then formal: $C' = DC^{(3)}D^{-1}$ by direct multiplication (the new coordinates are $f_0 = qg_2$, $f_1 = g_0$, $f_2 = -g_1$), $DS = DC^{(3)}D^{-1} = C'$, $SD = C^{(3)}$. Verified exactly on Y_{168} . $\square$

Corollary 3.5 (What the edge flow sees and what it does not). *Under the hypotheses of Theorem 3.3:*

- (i) $\ker \Psi \supseteq \{(f, -f, 0), ((q+1)f, 0, -f) : f = \omega^{k\tau} \mathbf{1}, k = 0, 1, 2\}$, a rank-6 sublattice: for the trivial pair and its two ω -conjugates the three edge functions are proportional, V_f is one-dimensional and carries only the eigenvalue $q^2 \omega^k$; hence $\text{rk } \Psi \leq 3n - 6$, with equality on Y_{168} (numeric rank 498).
- (ii) On Y_{168} the spectrum of L_E contains every pencil root except the six $\{1, \omega, \omega^2, q, q\omega, q\omega^2\}$; the remaining $|E| - (3n - 6) = 4n + 6$ eigenvalues all have modulus $q^{1/2}$ (numeric, to 10^{-4}): with $P(u) = \det(I - uC^{(3)})$,

$$\det(I - uL_E) \cdot (1 - u^3)(1 - q^3 u^3) = \det P(u) \cdot G(u),$$

$$G \in \mathbb{Z}[u], \quad \deg G = 4n + 6, \quad \text{all roots of } G \text{ on } |u| = q^{-1/2}.$$

- (iii) L_E is irreducible with row sums q^2 (its Perron eigenvalue), and 1 is not an eigenvalue of L_E (numeric, distance 0.58 to the nearest eigenvalue on Y_{168}).

Proof. (i) For $f = \omega^{k\tau} \mathbf{1}$: $f \circ t = \omega^{k\tau(x)}$, $f \circ h = \omega^{k(\tau(x)+1)}$, $m_f = (q+1)\omega^{k(\tau(x)+2)}$ on $(x \rightarrow y)$, proportional; the two displayed vectors are integral for $k = 0$, and over $\mathbb{Z}[\omega]$ for $k = 1, 2$, so the rational kernel has rank at least 6. (ii),(iii) are computations; the divisibility in (ii) is exact because $\det P(u)$ contains $(1 - u^3)(1 - q^3 u^3)(1 - q^6 u^3)$ and L_E has the eigenvalues $q^2 \omega^k$. $\square$

Remark 3.6 (Kang–Li context). Kang–Li [4] express the zeta function of a PGL_3 complex as a rational function in which the vertex factor $\det P(u)$ and a chamber (parahoric) factor appear; the operator L_E is the type-(1,0) gallery transfer operator whose determinant computes the zeta function counting closed type-raising geodesic galleries, and the modulus- $q^{1/2}$ remainder of Corollary 3.5(ii) is the parahoric spectrum, which for a Ramanujan complex lies on $|u| = q^{-1/2}$. We use nothing from [4]; the identity (ii) is established here numerically and its general form is left as it stands. The chamber flow L_F on oriented chambers $((a, b, c) \rightarrow (b, c, d), d \neq a)$, row sums q , has spectrum of moduli $\{1, q^{1/4}, q^{1/2}, q\}$ on Y_{168} and contains no nontrivial pencil root: the Hecke spectrum lives on edges, not on chambers.

Theorem 3.7 (The positive Hecke-dynamical algebra). *Let Y satisfy the hypotheses of Theorem 3.3 and let $\mathcal{O}_{L_E}$ be the Cuntz–Krieger algebra of L_E with its gauge action γ . Then $\mathcal{O}_{L_E}$ is a unital Kirchberg algebra whose gauge dynamics is the geodesic gallery flow; its partition function is $\det(I - uL_E)^{-1}$, whose poles include every nontrivial Satake datum $\lambda_{j,i}$ of the complex at $u = \lambda_{j,i}^{-1}$; its K -theory is $K_1(\mathcal{O}_{L_E}) = \ker(I - L_E^t) = 0$ and $K_0(\mathcal{O}_{L_E}) = \text{coker}(I - L_E^t)$ finite of order $|\det(I - L_E)|$ (on Y_{168} , $\log |\det(I - L_E)| = 603.249$, numeric). The integer intertwiner Ψ induces a $\mathbb{Z}[t^\pm]$ -module map $M_{C^{(3)}} \cong M_{C'} \rightarrow M_{L_E}$ of the Hecke module into the edge-flow module. $\mathcal{O}_{L_E}$ is not shift equivalent to $C^{(3)}$ (different nonzero spectra), and its K_0 is not the Hecke critical group $\mathbb{Z} \oplus \text{Jac}^H(Y)$.*

Proof. L_E is irreducible with row sums $q^2 \geq 4$ and is not a permutation matrix, so it satisfies condition (I) of Cuntz–Krieger and $\mathcal{O}_{L_E}$ is simple and purely infinite, separable, nuclear and in the UCT class, by the criteria quoted in [VI, Thm. 3.1]; purely infinite algebras are traceless. K -theory is the Cuntz–Krieger formula; $1 \notin \text{spec } L_E$ gives the kernel statement and finiteness. The module map is Theorem 3.3 and Proposition 3.4. The last sentence is Corollary 3.5(ii) and [X, Thm. 3.3]. $\square$

Proposition 3.8 (KMS states of the edge flow). *Let $\sigma_t(s_e) = e^{it}s_e$ be the gauge dynamics of $\mathcal{O}_{L_E}$. A KMS_β state exists if and only if $\beta = \beta_c := 2 \log q = \log \rho(L_E)$, and it is then unique: the state attached to the Perron eigenvector of L_E (the Parry measure of the gallery shift), which is the constant vector since the row sums are q^2 . The partition function $Z(\beta) = \sum_k \text{Tr}(L_E^k) e^{-k\beta} = -u \frac{d}{du} \log \det(I - uL_E) \big|_{u=e^{-\beta}}$ has poles at $u = \lambda^{-1}$ for every eigenvalue λ of L_E , hence at the inverses of all nontrivial Satake data by Theorem 3.3; the KMS state sees only $\lambda = q^2$. The Perron obstruction of [I, Thm. 2.5] therefore holds verbatim in rank two: the cuspidal GL_3 L -factors $P_j(u)^{-1}$ are carried by the partition function and by the KMS functionals obtained from the*

nontrivial eigenvectors of L_E (Theorem 3.3 gives them explicitly, $\Psi(f \circ t, f \circ h, m_f)$ -combinations), not by any KMS state.

Proof. For an irreducible non-negative matrix with the gauge dynamics, KMS_β states of the Cuntz–Krieger algebra correspond to non-negative eigenvectors of L_E^t for the eigenvalue e^β (as in [I, Thm. 2.4]); by Perron–Frobenius the only such eigenvalue is $\rho(L_E) = q^2$ (row sums), with a unique normalized positive eigenvector. The rest is Theorem 3.3 and the identity $\text{Tr}(L_E^k) = \sum_\lambda \lambda^k$. $\square$

Remark 3.9 (Bulk and boundary, in rank two). [I, Thm. 3.7] found that the natural positive system of a graph — the boundary algebra $\mathcal{O}_{\text{Anb}}$ — has K -theory blind to the Hecke data, while the Hecke critical group $\mathbb{Z} \oplus \text{Jac}(Y)$ lives in the Laplacian channel; [VI] realized the latter by a Hecke-functorial, not Hecke-dynamical, Kirchberg algebra, and [II] showed no single algebra can be both. In rank two the same dichotomy appears with one difference: the positive system $\mathcal{O}_{L_E}$ carries the whole nontrivial Hecke spectrum (Theorem 3.3), and its K_0 is a finite group unrelated to $\text{Jac}^H(Y)$, while $\mathcal{O}(Y, \Delta^H)$ of [X] carries $\mathbb{Z} \oplus \text{Jac}^H(Y)$ without dynamics. An algebra carrying both would be $\mathcal{O}_B$ for a non-negative $B \sim_{\mathbb{Z}} C^{(3)}$; [II, Thm. 1.6] no longer forbids it (there is no trace obstruction, [X, Thm. 4.2]), and its existence is the open instance of the weak Generalized Spectral Conjecture of Remark 2.4. That is the exact residue of the problem.

4. THE VERIFICATION BATTERY

check	complexes	result
(G) $\deg = i - \tau$ makes $C^{(3)}$ degree one; $P_d = C^{(3)3} _d$; $\text{spec } P_0 = \Lambda_Q$; $\det P_0 = q^{3n}$	$Y_{21}, Y_{24}, Y_{42}, Y_{168}$	exact (spectrum numeric)
(Y) P_0 : negative entries 147, 184, 476, 5936; right Perron > 0 ; left Perron $\not\geq 0$; no $P_0^k > 0$, $k \leq 30$; negative digraph cyclic	same	exact / numeric
(N) LP relaxation $Y \geq -P_0$, $Y P_0 \leq 0$, $\text{Tr } Y = 0$ infeasible; exact Farkas certificates (denominators 2, 128)	Y_{21}, Y_{24} (Y_{12} float)	exact
(L) cyclic lift of the trivial shift equivalence: four equations, lag 3	four complexes	exact
(E) L_E : 1176 edges, row sums $4 = q^2$, irreducible; 498 of 504 pencil roots present; missing $\{1, \omega, \omega^2, 2, 2\omega, 2\omega^2\}$; remainder 678 = $4n + 6$ of modulus $\sqrt{2}$	Y_{168}	numeric
(I) $L_E \Psi = \Psi C'$; $\text{rk } \Psi = 498 = 3n - 6$; $D, C^{(3)} D^{-1}$ integral; lag-one equations	Y_{168}	exact
(T) $\text{Tr } L_E^k$ vs $\text{Tr}(C^{(3)})^k$, $k \leq 10$: unequal (parahoric spectrum)	Y_{168}	exact
(F) L_F : 3528 oriented chambers, row sums 2, irreducible; spectrum moduli $\{1, 2^{1/4}, 2^{1/2}, 2\}$; 3 pencil roots	Y_{168}	numeric

TABLE 1. Verification (`pgl3_dynamical.py`). Y_{168} is the only computed complex satisfying the flag condition; the edge-flow theorem is checked there. $\log |\det(I - L_E)| = 603.249$ on Y_{168} .

5. SCOPE, RESIDUE, CORRECTIONS

Remark 5.1 (Solved, open, impossible). *Solved:* the graded structure of $C^{(3)}$ and the reduction of its realization problem to the primitive-matrix problem for the $n \times n$ return map, with an explicit lag formula; the module criterion and the cofactor construction; the impossibility of the nilpotent-cofactor route on Y_{21}, Y_{24} (exact certificates); the geodesic edge flow theorem with its explicit intertwiner, giving a non-negative integer matrix whose spectrum contains the entire nontrivial PGL_3 Hecke spectrum, and the Kirchberg algebra $\mathcal{O}_{L_E}$ it generates. *Open:* (a) $C^{(3)} \sim_{\mathbb{Z}} B \geq 0$, an instance of the weak Generalized Spectral Conjecture over $\mathbb{Z}$ (Remark 2.4); (b) a proof of the determinant identity of Corollary 3.5(ii) for all $\tilde{A}_2$ complexes, and the identification of $G(u)$ with Kang–Li’s chamber factor; (c) the structure of the finite group $K_0(\mathcal{O}_{L_E})$; (d) whether positive presentations of M_{P_0} of higher degree $K \geq 2$ exist (Proposition 2.2) — the Farkas method extends but the search space grows. *Impossible or false as stated:* the “century deadlock since Bost–Connes” that positive correspondences cannot carry higher automorphic Hecke spectra — Theorem 3.3 carries it, as Hashimoto’s matrix already did in rank one; what positive systems have so far failed to carry is the Hecke *module* (the K -theory), and whether they can is the open conjecture.

Remark 5.2 (Corrections to the task specification). (1) The specification asked for explicit (B, R, S, ℓ) with $B \geq 0$ shift equivalent over $\mathbb{Z}$ to $C^{(3)}$, as a corollary of [X]’s “Boyle–Handelman spectral conditions”. Those conditions give the nonzero spectrum (Spectral Conjecture, a theorem over $\mathbb{Z}$ by [5]); shift equivalence is the weak Generalized Spectral Conjecture, open. No such (B, R, S, ℓ) is supplied, and none could be without proving a case of that conjecture. (2) “Method one” — a block-cyclic B whose three-step product cancels $-qA_2$ — is exactly Lemma 1.3: correct as a reduction, and it converts the problem into the primitive-matrix problem for P_0 rather than solving it. (3) “Method two” (state splitting) applies to non-negative matrices; it cannot start from $C^{(3)}$, and the algebraic substitute (cofactors, Proposition 2.2) is obstructed at degree two (Theorem 2.3). (4) The genuine positive GL_3 dynamics is the geodesic edge flow L_E , whose existence and spectral content are proved here; it is the rank-two Hashimoto matrix, not a realization of $C^{(3)}$. (5) The claim that KMS states at the critical temperature “reproduce the GL_3 local L -factors” is false in the same way it was false in rank one: the unique KMS state sees the Perron eigenvalue only, the L -factors live in the partition function (Proposition 3.8); the “curse” of [II] is not broken but located — it concerns the Hecke module, not the Hecke spectrum. (6) Of the specification’s four checks, $B \geq 0$, irreducibility and the spectral containment hold for L_E ; the four shift-equivalence identities hold for the intertwiner pair $(D, C^{(3)}D^{-1})$ between $C^{(3)}$ and C' (lag one) and for Ψ as a one-sided intertwiner into L_E . The trace identities $\mathrm{Tr} L_E^k = \mathrm{Tr}(C^{(3)})^k$ fail, and so does the Smith-form identity $\mathrm{coker}(I - L_E) \cong \mathbb{Z} \oplus \mathrm{Jac}^H$; both must fail, because L_E carries the additional parahoric spectrum (Table 1, line (T)).

ACKNOWLEDGEMENT OF PROVENANCE AND CAVEATS

Unrefereed preprint. Every numbered statement is proved except the items marked numeric in Table 1; Theorem 2.3 is certified by exact rational arithmetic. The statement of the weak Generalized Spectral Conjecture and its status are taken from the introduction of [3] (consulted), which attributes the weak form to [1, p. 253] and [2, p. 124]; those two pages were not consulted directly. The identification of the thick complexes with quotients of the building of PGL_3 over a specific field is not claimed. Bibliographic data of [1, 2, 3, 5, 4] were checked against publisher or arXiv pages.

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